

Let's Talk about Statistics and Probability

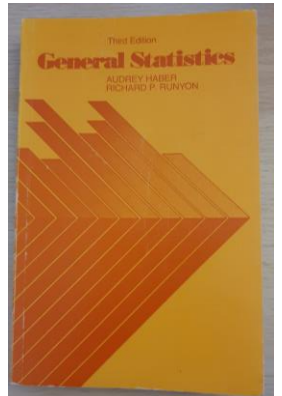
Statistics: Samples and Populations
Probabilities in action

What is/are statistics?

Two definitions offered:

1. “Statistics is commonly regarded as a ***collection*** of numerical facts which are expressed in terms of summarizing statements and which have been collected from several observations or from other numerical data.”
2. “Statistics may also be regarded as a ***method*** of dealing with data. This definition stresses the view that statistics is a tool concerned with the collection, organising and analysis of numerical facts or observations.”

Haber, Audrey, and Richard P.
Runyon. *General statistics*. 1973.



Working with Real life data

There are 2 kinds of people

1) Those who can extrapolate with missing data

The Difference between Raw Data & the Stories Data can tell

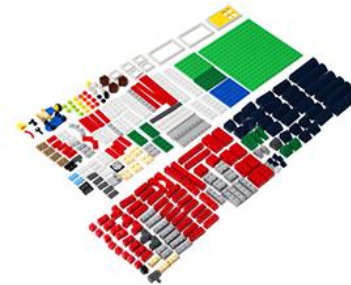
1 Data
Collection



2 Data
Preparation



3 Data
Visualization



4 Data
Analysis



5 Data
Storytelling



www.effectivedatastorytelling.com

Statistics have power

“ On average, we have reduced the carbon impacts at our sites by 10%”

Site	Carbon Reduction
Site A	0%
Site B	0%
Site C	0%
Site D	0%
Site E	0%
Site F	0%
Site G	0%
Site H	0%
Site I	-50%
Site J	-50%

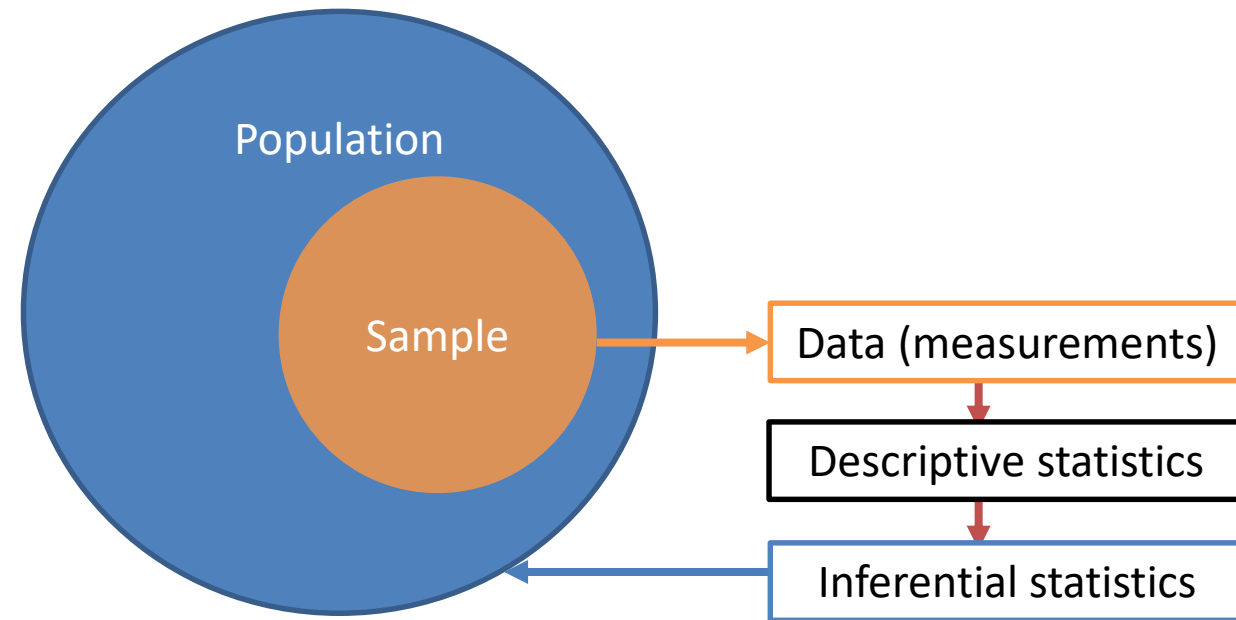
“ On average, we have increased salaries by 10%”

Site	Salary Increase
Employee A	2%
Employee B	2%
Employee C	2%
Employee D	2%
Employee E	2%
Manager A	4%
Manager B	4%
Manager C	4%
Director A	40%
Director B	40%

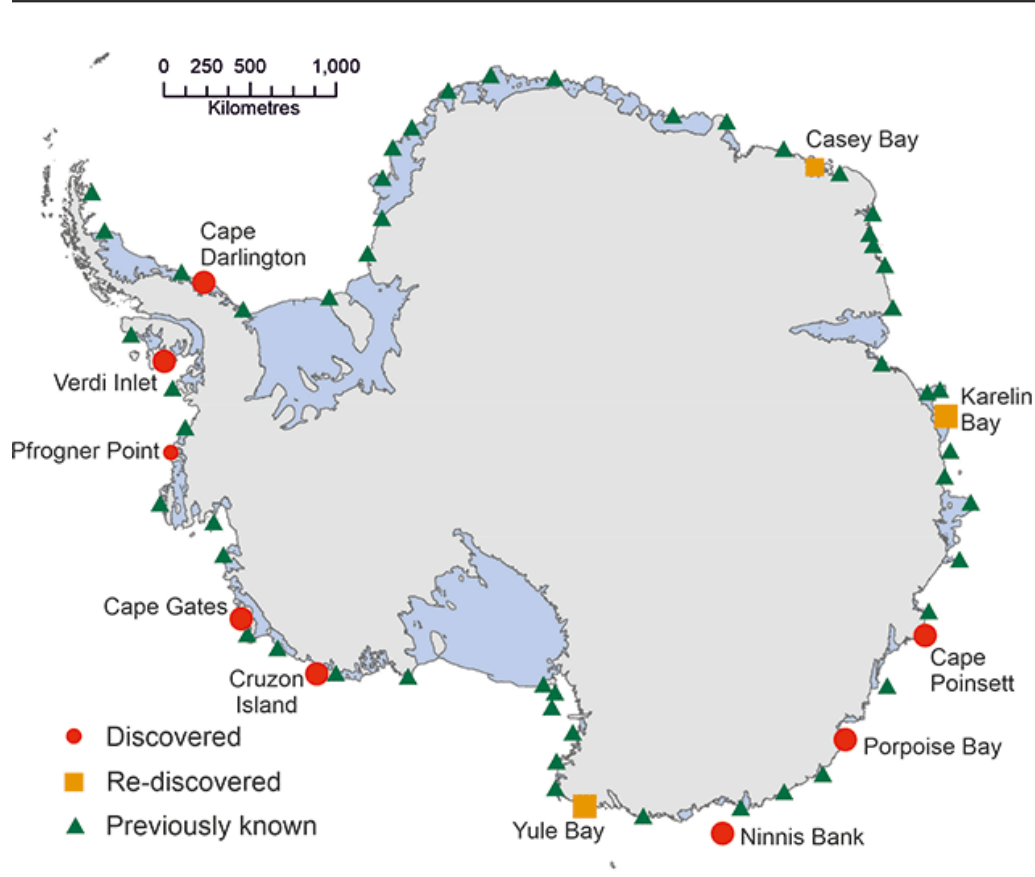
SAMPLES AND POPULATIONS

Samples

- Use samples when:
 - Population is too large or too complex to look at it all
 - Learn what we can from a sample and then build a **model** for, or **infer** to the larger population
- Samples need to be representative
 - **Spread**
 - **Size**



Estimating the number of Emperor Penguins



Data (measurements)

- Count the **number of colonies**
- Measure the size of huddles in each of the colonies

Descriptive statistics

- Count penguins in a **SAMPLE** of colonies
- Infer the number of penguins in a sized huddle

Inferential statistics

- Estimate the total number of penguins**

Be careful with statistics

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Science & Environment

Climate change: Satellites find **new** colonies of Emperor penguins

By Jonathan Amos
BBC Science Correspondent

5 August 2020

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Scientists have just discovered **11 new** emperor penguin colonies in Antarctica

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A space-based sensor has detected new colonies of emperor penguins on Antarctic sea ice. Credit: Christopher Walton

ECOLOGY · 07 AUGUST 2020

Satellites find penguins by following the poo

Images from space bolster the population count, but the birds remain vulnerable to climate change.

BUSINESS INSIDER

11 new colonies of emperor penguins have been discovered in Antarctica with satellite images

Haven Orecchio-Egredit Aug 6, 2020, 5:08 PM



A discovery of 11 emperor penguin colonies

SCI NEWS

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Eight New Colonies of Emperor Penguins Discovered in Antarctica

Aug 12, 2020 by News Staff / Source

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ANIMAL

Emperor Penguin **Populations are Increasing** in Antarctica!

5 months ago

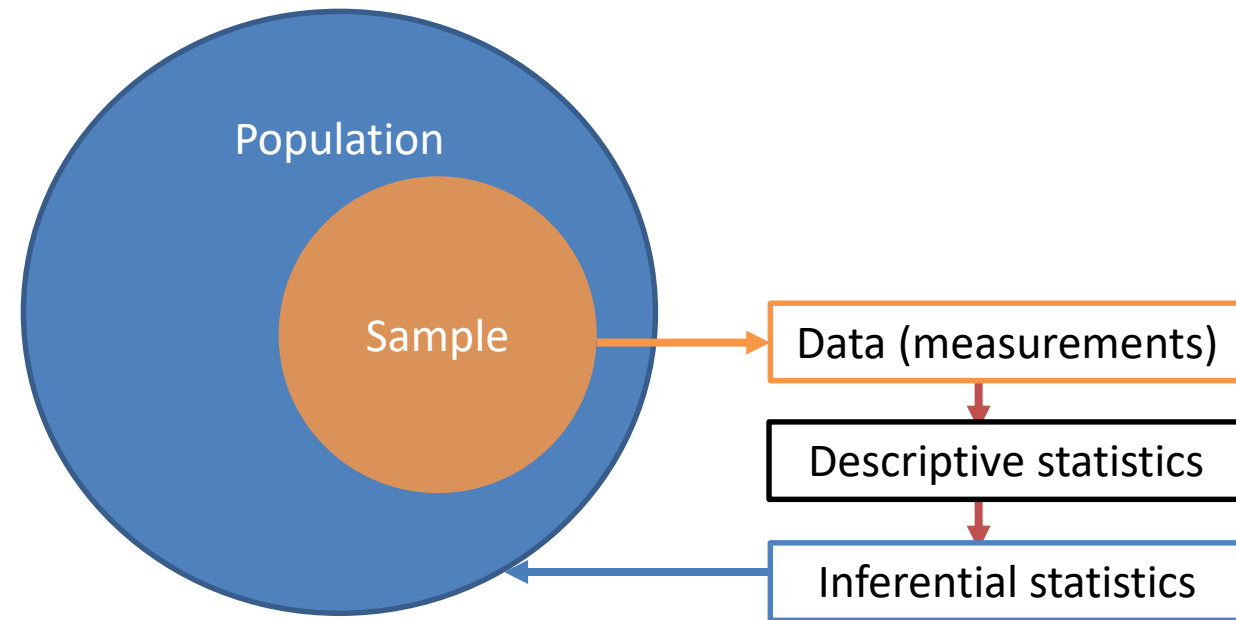
By Eliza Erskine

DEPARTMENT OF ELECTRONICS

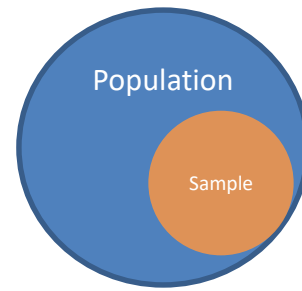
Save

Let's take a look at samples again

- Use samples when:
 - Population is too large or too complex to look at it all
 - Learn what we can from a sample and then build a **model** for, or **infer** to the larger population
- Samples need to be representative
 - **Spread**
 - **Size**



UK Domestic Demand Data



The more data we have of demand, the better we can forecast it.

Ideally:

- A smart meter in every home so we can instantaneously measure all demand
- But, we're not there yet
- So how can we make estimates?

Data (measurements)

↓
Descriptive statistics

↓
Inferential statistics

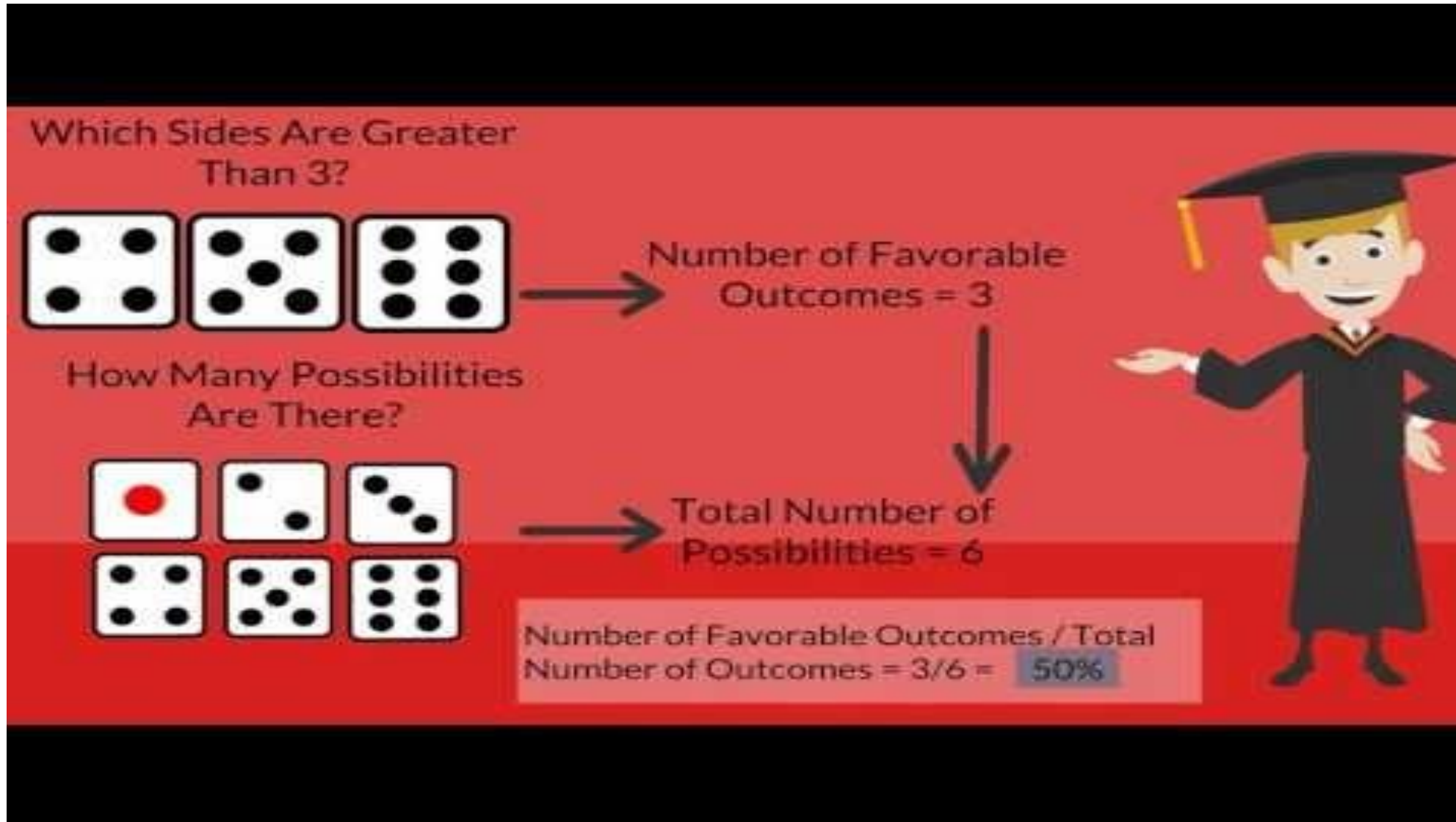
Let's get thinking about

PROBABILITIES

Introduction

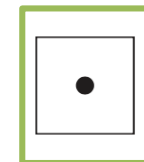
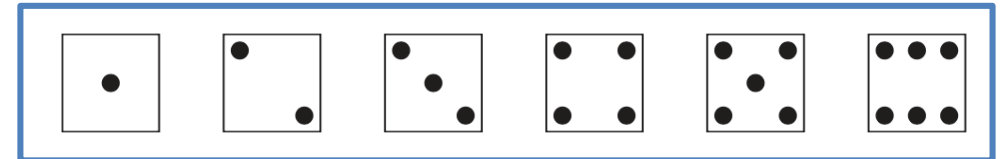
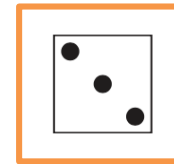
- When we make decisions day to day we decide what actions will **likely** lead to success
- When we play games of chance we often have a feeling of what is **probably** going to happen next
- We use our *intuition*, past experience and '*gut feeling*' to know what is most likely to occur
- We need consider more empirically though of the probabilities involved to make sure that our judgements are sound

Probability



Getting the language right

- **Experiment** (e.g. Rolling a die)
 - *Experiment* and *trial* are used interchangeable
- The result of an experiment is called an **outcome**
- The set of *all* possible outcomes of a probability experiment is called the **sample space**
- Any *subset* of the sample space is known as an **event** (*a single or set of possible outcomes of the experiment*)

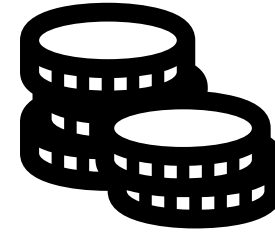


Roll of 1



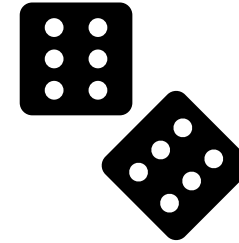
Roll greater than 3

Probability – Coin Toss



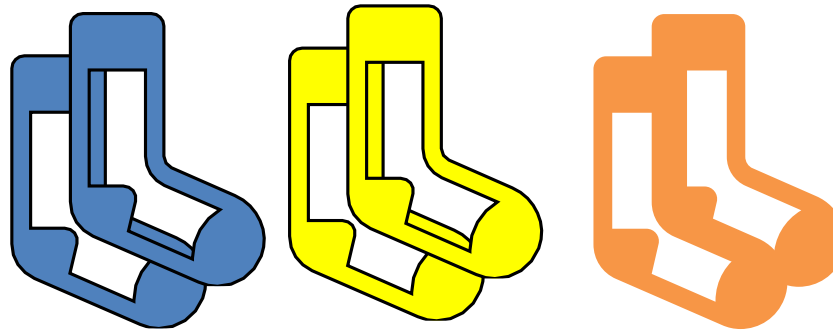
- 2 **outcomes**, Heads or Tails, each with $1/2$ Probability
- The **sample space** {Heads, Tails}
- An **event**: If I toss a coin 5 times what is the most likely result? **Poll!**

Probability – Roll of a dice



- 6 **outcomes**, 1 or 2 or 3 or 4 or 5 or 6, each with 1/6 Probability
- The **sample space** {1,2,3,4,5,6}
- Side greater than three?
 - An **event**: 3 favourable outcomes (4,5,6)
 - The **sample space**: 6 possibilities (1,2,3,4,5,6).
 - Probability $\left(\frac{N_{fav.outcomes}}{N_{sample\ space}}\right)$ is 3/6 or 1/2
- If I roll 6 times am I guaranteed a 3 exactly once? **Show of Hands!**
 - (Gamblers Fallacy)

Probability – Sock Pull



- In your sock drawer there are 2 blue socks, 2 orange socks and 2 yellow socks, distributed so that the probability of picking each is equal.
- Without looking, you draw one at put it to one side before drawing another at random.
- What is the probability that the two socks will be the same colour? **Poll!**
 - Now we are dealing with a set that is not independent.
 - Our first action (pulling out sock #1) has an impact on the probability of our second action (pulling out sock #2)

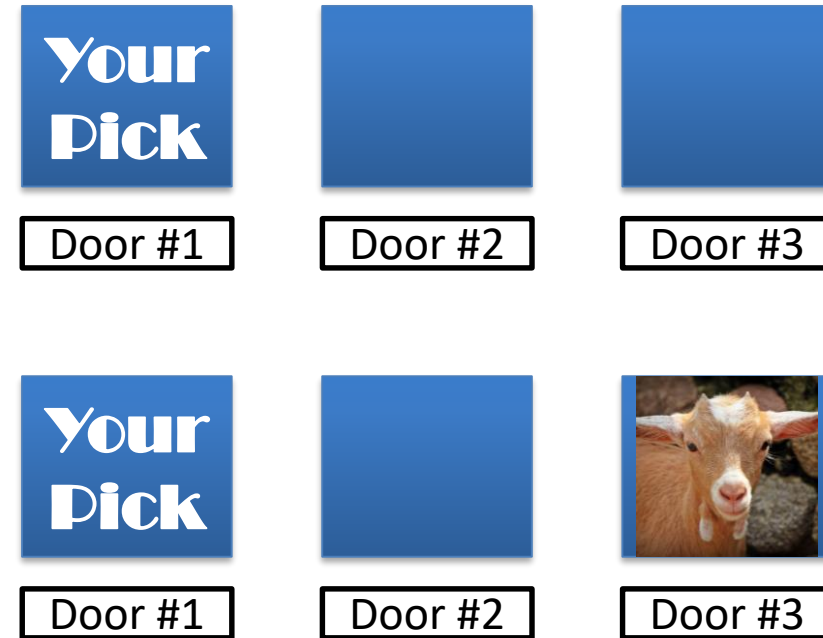
The Monty Hall problem

- You are presented with three doors
- One door has behind it a car, the other two goats



The Monty Hall problem

- You pick a door (say number 1)
- Monty shows you that there is a goat behind one of the other doors
- The choice now is simple:
 - Stick with your original Choice
 - Switch to the alternative



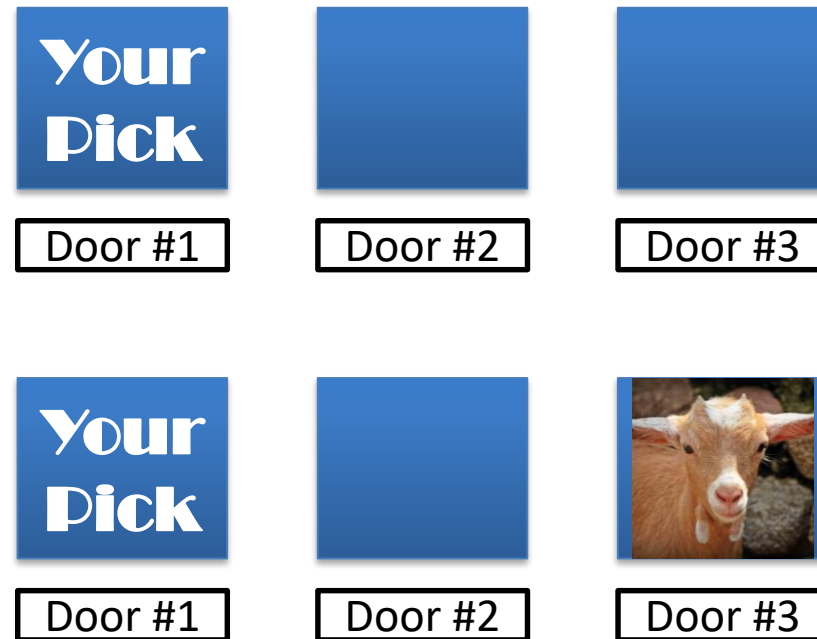
You Decide!

Poll!

Discussion – Online Simulations & think empirically

- Build a small group of people who don't all agree, discuss and try to convince each other...

- Initially the probability of winning is $1/3$
- Once you are given the new information, does the probability of winning increase, decrease or stay the same if you stick/switch?



Poll!

Conclusion

Statistics

- Statistics have power, use them wisely!

Probabilities

- Probability is a ‘measure of the plausibility of a particular outcome’ – a variable being observed taking particular value(s)
 - It’s important we consider all possible outcomes – probabilities are linked to the set of possibilities
 - It’s important to take account of dependence and independence
- Need to be careful that ‘gut feelings’ do not distract us – think empirically and use the mathematical techniques you have learnt
 - When considering failure rates of electrical components, likelihood protection fails or the certainty that results are ‘correct’